DATA 266 — Homework 11

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SID4 = 0670 | SEED = 670 | SLICE = 670 | HP_ID = 4 | CLS_A = 0 | CLS_B = 5

Notebook: `clip_cifar.ipynb`. Model: `openai/clip-vit-base-patch32`. Dataset: CIFAR-10.

HP_ID is reported only.

1. Setup

Item	Value
Model	`openai/clip-vit-base-patch32` (~151M params)
Dataset	CIFAR-10 (train for Part A, test for Part B)
Caption	`a photo of a dog`
Part A sample	100 images per class (1,000 total), SEED=670
Zero-shot prompts	`a photo of a {class}` for all 10 classes
Device	MPS
Batch size	64

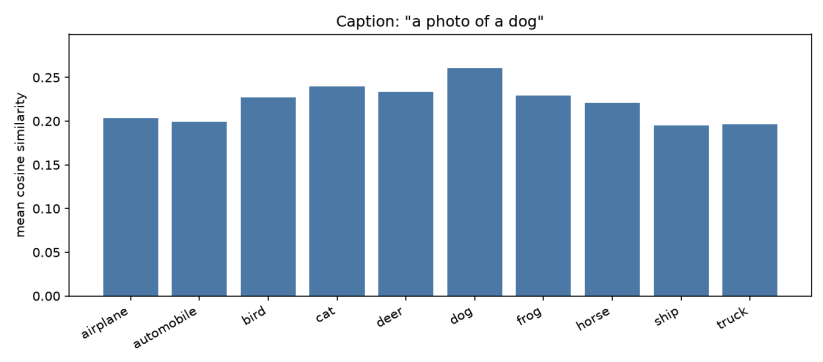
I load CLIP with the matching processor. Text and image embeddings are L2-normalized, then compared with cosine similarity (dot product of unit vectors).

2. Caption–image similarity

I scored the caption against 1,000 CIFAR-10 train images (100 per class).

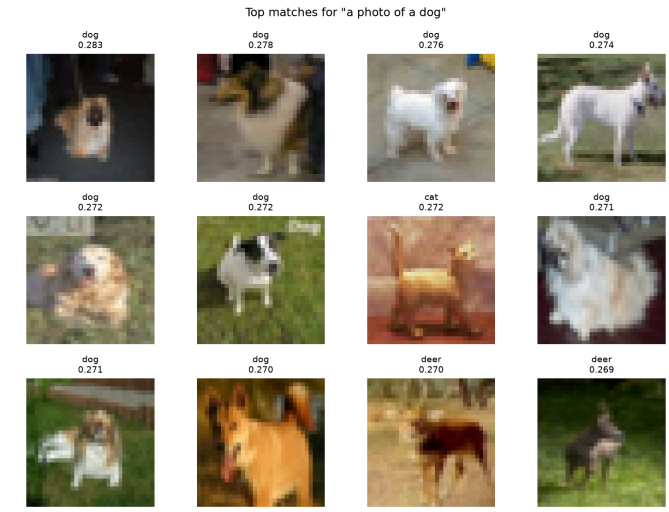
Class	Mean similarity
airplane	0.204
automobile	0.200
bird	0.227
cat	0.240
deer	0.234
dog	**0.260**
frog	0.229
horse	0.221
ship	0.195
truck	0.196

Highest mean class: **dog** (0.260).



Mean caption similarity by class

Dog ranks first, then cat and other animals. Vehicles sit at the bottom — the expected order for a dog caption.



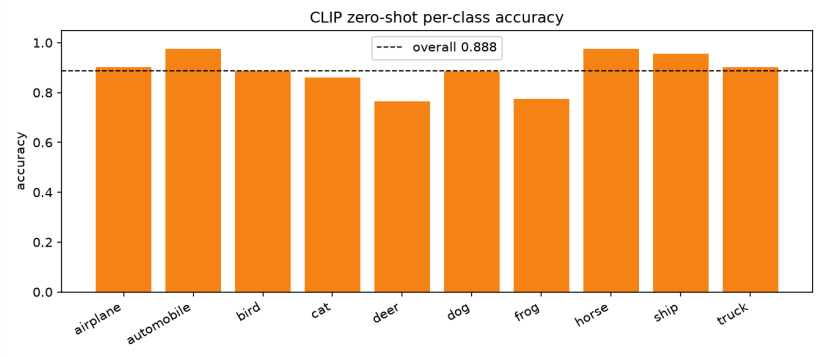
Top matches for the caption

The top-12 gallery is mostly dogs, with a few other animals that still look somewhat dog-like at 32×32.

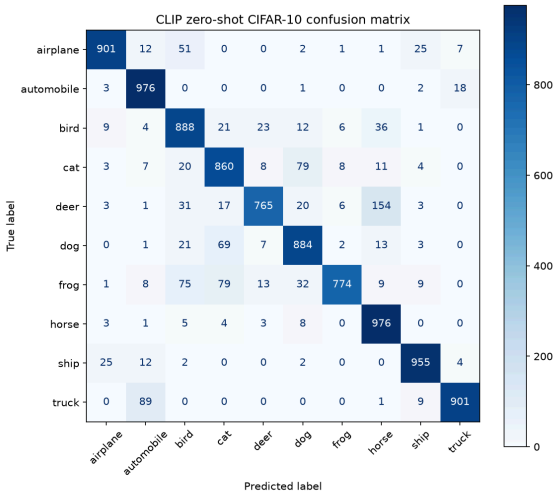
3. Zero-shot CIFAR-10 classification

I embedded all 10 class prompts and classified every CIFAR-10 test image (n=10,000) by the highest cosine similarity. Overall accuracy: **0.888**.

Class	Accuracy
airplane	0.901
automobile	0.976
bird	0.888
cat	0.860
deer	0.765
dog	0.884
frog	0.774
horse	0.976
ship	0.955
truck	0.901



Per-class zero-shot accuracy



Confusion matrix

Automobile and horse are strongest (97.6%). Deer is weakest (76.5%), mostly confused with horse. Cat↔dog and frog↔bird/cat swaps show up too — expected for similar animals at CIFAR resolution.

Conclusion

CLIP ranked the dog caption highest on dog images with no fine-tuning. Zero-shot CIFAR-10 reached 88.8% accuracy. Errors concentrate on visually similar pairs (deer/horse, cat/dog), matching the confusion matrix.